

Endpoint profiles on their natural scale

Complete theorem; independently reviewed

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Theorem 1. *Under the monotone mixed-moment hypotheses of `optimal_interior_endpoint_profile.tex`, put $u_i = 1 - p_i$ and $s = \sum_i u_i$. For every fixed $L < \infty$, uniformly over rows with $0 \leq s \leq L$,*

$$\max_i |mu_i - s| \leq C_L \left(\sqrt{\frac{s}{m}} + \frac{1}{m} \right).$$

The two inputs beyond the reviewed fixed-annulus profile theorem are the sharp relative balance and endpoint distribution bound in `./sharp_endpoint_relative_balance.tex`, and the variable-center Jackson estimates in `./rational_designs/asymptotic/variable_jackson_endpoint_kernel.tex`. The present note proves the measure and monotonicity assembly.

Coordinates and sharp bounds. Retain $M = m - 1$, $\ell = \lfloor M/4 \rfloor$, $N = M - \ell$, $z_i = \ell u_i$, and the marked row probabilities

$$x_i = M^{-1} \sum_{j \neq i} z_j, \quad h_i = z_i - x_i, \quad d\mu_i = (\ell + 1) \mathbb{E}_B \prod_{j \in B} p_j dw.$$

Here B ranges over the ℓ -subsets of the unmarked indices. The reviewed sharp bounds give, uniformly in i ,

$$\max_j z_j \leq C(x_i + m^{-1}), \quad \mu_i([0, t]) \leq C(t + m^{-1}) \quad (t \geq 0). \quad (1)$$

For the first bound, absorb the marked column's contribution in the full-family balance estimate. For the second apply the sharp CDF lemma to the deleted M -column tensor and note that $\mu_i \leq CMw$, with only a fixed rescaling of its deficit coordinate. Also μ_i has total mass one.

For centers $a \geq T/m$, with T a sufficiently large fixed constant, write

$$b_a = \sqrt{a/m}, \quad E_a(x) = b_a^{-1} (1 + |x - a|/b_a)^{-2r} \quad \text{when } x \in [ca, Ca].$$

Fixed changes in the constants or scale of this envelope are harmless. The Jackson degree is a sufficiently small fixed multiple of m , and its fixed power $r > 4$ can be increased for the negligible global high-order remainder. Let Q_a be the corresponding positive polynomial, normalized so its scalar beta-reference integral lies between two positive absolute constants. Its near-diagonal lower bound is $Q_a \geq c/b_a$ on $|x - a| \leq cb_a$.

The variable localized comparisons. For $T/m \leq a \leq a_*$, where $a_* > 0$ is fixed and sufficiently small, the variable-kernel note gives, with $H = \max_j z_j - \min_j z_j$,

$$|\mu_i(Q_a) - \beta_\ell(Q_a)| \leq \frac{C}{a} \int_{x_i \in [ca, Ca]} H^2 E_a(x_i) d\mu_i + C(ma)^{1/2-r} + m^{-A}, \quad (2)$$

$$|\mu_i(h_i Q_a)| \leq C \int_{x_i \in [ca, Ca]} \left(\frac{H^2}{\sqrt{ma}} + \frac{H^3}{a} \right) E_a(x_i) d\mu_i + Ca(ma)^{1/2-r} + m^{-A}. \quad (3)$$

Here $A > 10$ is fixed, and the constants are independent of the varying center. The same relative tails bound scalar kernel integrals outside the displayed relative annulus, including an extra factor $|h_i|$ in the marked case. These follow from (1), not from the total-mass bound alone. On the relative annulus, (1) gives $H \leq Ca$ because $ma \geq T$.

Lemma 1 (Uniform variable-width local mass). *After choosing T sufficiently large and then a sufficiently small fixed $a_0 > 0$, uniformly for $T/m \leq a \leq a_0$ and all marked columns,*

$$\mu_i(|x_i - a| \leq b_a) \leq Cb_a, \quad c \leq \mu_i(Q_a) \leq C, \quad \int_{x_i \in [ca, Ca]} E_a(x_i) d\mu_i \leq C.$$

Moreover, the portion of the last integral outside $|x_i - a| \leq Bb_a$ is at most CB^{1-2r} , up to the separately controlled relative-annulus tails.

Proof. First prove only the upper bounds. Choose a fixed small a_1 for which the coefficient Ca in (2) is small enough for absorption on $a \leq a_1$. Consider the supremum

$$V = \sup_{i, T/m \leq a \leq a_1} \mu_i([a - cb_a, a + cb_a])/b_a.$$

For centers below T/m , (1) bounds this normalized local mass by a fixed constant depending on T , using width $\sqrt{a/m} + 1/m$. For centers in a fixed interval above a_1 , the already proved fixed-annulus mass bound supplies a finite uniform constant. Thus, for any center in the supremum, intervals used to cover $[ca, Ca]$ have normalized local mass at most $V + C_T$; here C_T can also depend on the fixed upper boundary a_1 . Widths on this relative annulus are comparable to b_a . Summing the polynomial envelope over these intervals gives

$$\int_{[ca, Ca]} E_a d\mu_i \leq C(V + C_T).$$

The lower bound for Q_a on its central window, its bounded reference integral, and (2) now imply

$$V \leq C + Ca_1(V + C_T) + CT^{1/2-r}.$$

Absorb Ca_1V . This proves a finite upper bound $V \leq C_1$, uniform in m and the marked column. It gives the stated upper interval and envelope bounds for all centers below a_1 .

Now choose T larger, if necessary, so the relative tail term in (2) is smaller than a fixed fraction of the positive reference mass. The preceding upper bound remains finite with this choice. Choose $a_0 \leq a_1$ still smaller so that Ca_0C_1 is also smaller than that fraction. Equation (2) proves the lower kernel-mass bound for $T/m \leq a \leq a_0$. The tail estimate follows by summing the local interval bounds over length- b_a intervals. This two-stage choice avoids assuming that fixed-annulus constants remain bounded when its lower endpoint tends to zero. $\square$

A normalized maximum. Use the common coordinate and deviations

$$y = \ell s/m, \quad h_i^* = z_i - y, \quad x_i = y - h_i^*/M, \quad h_i = (m/M)h_i^*.$$

Put $b(y) = \sqrt{y/m} + 1/m$. Sharp balance gives

$$|h_i^*| \leq C(y + 1/m), \quad |x_i - y| \leq C(y + 1/m)/m = o(b(y)) \quad (4)$$

uniformly on bounded y -intervals. The same bounds hold for any row entering the corresponding deleted-coordinate interval, by absorbing the marked contribution as in (1).

Fix a bounded y -interval containing the desired range and a fixed enlargement. Consider the maximum

$$D = \max_{i, \text{rows in that interval}} |h_i^*(y)|/b(y).$$

Rows with my bounded have bounded normalized deviation by (4). Rows with y bounded below by any fixed positive constant have bounded normalized deviation by the already proved interior square-root theorem. Consequently, if D exceeds a sufficiently large fixed constant, a maximizing row y_0 lies in

$$2T/m \leq y_0 \leq a_2,$$

where $a_2 > 0$ can be chosen as small as needed after Lemma 1. Write $\Delta = |h_i^*(y_0)|$ and $b_0 = b(y_0)$. Then $\Delta = Db_0$ and $\Delta \leq Cy_0$. For rows whose coordinates lie in a fixed relative neighborhood of y_0 , maximality gives

$$|h_j^*| \leq C\Delta \quad \text{for every } j, \quad H \leq C\Delta. \quad (5)$$

This also holds in the relative neighborhood of the selected deleted coordinate x_0 , by (4).

A favorable one-sided window. Choose fixed B_0 sufficiently large and $A_0 = 2B_0$. Center the test at $a = x_0 + A_0b_0$ for a positive selected deviation, and at $a = x_0 - A_0b_0$ for a negative one. Since $\Delta = Db_0 \leq Cy_0$, one has $y_0/b_0 \geq D/C$. Choosing the fixed contradiction threshold for D after B_0 therefore keeps these centers and windows in a fixed relative neighborhood of y_0 , without changing the already fixed T . Then $a \asymp y_0$ and $b_a \asymp b_0$. The favorable window lies strictly on the selected side, so monotonicity gives the chosen sign of h_i with magnitude at least $\Delta/2$ there, whenever D is sufficiently large.

Lemma 1 and its local tails imply a positive kernel mass on that favorable window. Within the relative annulus, (5) bounds the opposite-sign contribution outside it by $C\Delta B_0^{1-2r}$, made small by choosing B_0 large. Outside the relative annulus the scalar marked tail is at most

$$Ca(ma)^{1/2-r} = Cb_a(ma)^{1-r}.$$

Since $ma \geq T$ and $\Delta \asymp Db_a$, this is a small fraction of Δ after the fixed choices. Hence

$$|\mu_i(h_i Q_a)| \geq c\Delta. \quad (6)$$

The opposite bound (3), the local envelope mass estimate, and (5) give

$$|\mu_i(h_i Q_a)| \leq C \left(\frac{\Delta^2}{\sqrt{ma}} + \frac{\Delta^3}{a} \right) + Cb_a(ma)^{1-r} + m^{-A}.$$

Since $\Delta \leq Ca$ and $a \leq Ca_2$, the ratio of the first part to Δ is at most $C\sqrt{a/m} + Ca$. Choose a_2 sufficiently small and then m large. The tail is absorbed using $ma \geq T$ and the choice of D above the fixed threshold. Finally $\Delta \geq c/m$, so m^{-A} is negligible. This contradicts (6). Therefore D is bounded by a constant depending only on the desired upper endpoint L . The relation $mu_i - s = (m/\ell)h_i^*$ and $y \asymp s$ proves Theorem 1.

Relative synchronization and the opposite endpoint. For $s > 0$, the theorem gives

$$\max_i \left| \frac{mu_i}{s} - 1 \right| \leq C_L \left((ms)^{-1/2} + (ms)^{-1} \right).$$

Thus the scaled columns synchronize relatively whenever $ms \rightarrow \infty$ and $s \leq L$. Reversing the row order and replacing every p_i by $1-p_i$ preserves the mixed moments, since $\int_0^1 (1-t)^r dt = 1/(r+1)$. The same conclusion therefore holds at the opposite endpoint, with s replaced by $\sum_i p_i$ and u_i by p_i .

The two orders in the theorem are necessary in this generality. The reviewed notes `../endpoint_profile_fluctuation_barrier.tex` and `../finite_profile_realization.tex` give fluctuations of order $m^{-1/2}$ at fixed positive s . The note `../endpoint_profile_sharpness.tex` gives actual finite uniform dice with $s \asymp m^{-2}$ and error $\asymp m^{-1}$, so the additive floor cannot be removed. The subsequent note `intermediate_endpoint_sharpness.tex` gives matching finite examples at every positive bounded scale for the refined envelope $\min\{ms, \sqrt{s/m} + 1/m\}$, using the trivial upper bound ms . The realized deficit is comparable to the prescribed scale; no exact prescribed numerical value of s is asserted.